

A Comparison of the Animal Recognition between the Real Objects and the Modeled 3D Objects

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ABSTRACT

This paper proposed about comparison of the animal recognition processed on the real world objects and the modeled 3D objects. The object classification consists of the three main steps – feature extraction, training classifier and image evaluation. Feature description is based on locale visual descriptors like SIFT or SURF. Support Vector Machine (SVM) in combination with the bags of visual keypoints (BOW) is used to classify descriptors. The 3D models were created using photogrammetry and 3D modeling for ideal 3D models of mammals. This method of 3D model creation is described in details.

Index Terms— Animal recognition, 3D model, Support Vector Machine

1. INTRODUCTION

In the computer vision, the main idea of object recognition is creation a representation of the objects from particular classes, which characterized appearance of these objects and in this way determine the class for the unknown object [1, 2]. In this paper, we are focusing on the recognition of the big mammal's occurring in Slovak republic. There were chosen 5 classes for the recognition, namely wolf, wild boar, deer, brown bear and fox.

To develop a system, which is capable of acceptable mammal's recognition, the good training and testing data were required. Collection and creation of these databases in the real world is very expensive and time consuming, because of the natural mammals behavior. In this paper, a new method of the testing data creation was presented. Instead of capturing the testing data in the wild countryside, these data was created using 3D object modeling methods. Then, the recognition results on the 3D modeled testing data were compared with the real world testing data.

The rest of the paper is organized as follows: in the second section, the method of the 3D objects modeling is described. In the third section, object recognition process is presented and the description of the classification model creation follows in the next section. The experimental results are given in the fifth section. In the end, conclusion and future work are presented.

2. 3D OBJECT MODELING

The 3D models using combination of photogrammetry and 3D modeling for ideal 3D models of mammals were created. The true copy of the original texture for the real-looking 3D model was created using the photogrammetry. The models were then optimized and this resulted in the reduction of the number of the polygons. After this step, the model symmetry was adjusted and the model was rotated into the correct direction. In the last step, the texture of the optimized model was corrected [3]. The example of 3D object creation is shown in Fig. 1.

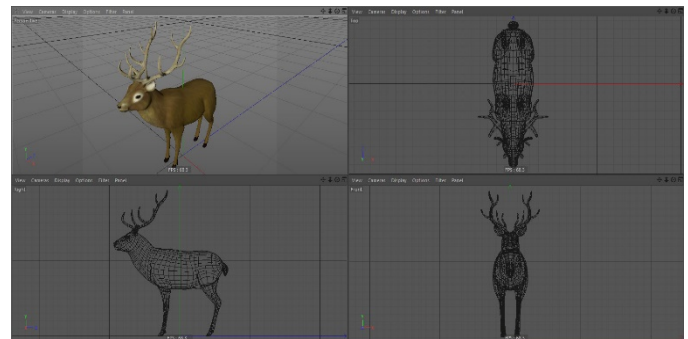


Fig. 1. The example of 3D object creation

2.1. The generation of the 3D model

The first step in the procedure of 3D object creation was taking the photos of the mounted animals from the different angles and positions. Some part of the animals were pictured in the more detail to increase the model precision. For every model, more than 150 pictures were taken. The 3D model mask using these pictures was created. Afterwards, 3D models with texture in the high resolution were exported. The optimization process of the 3D object is shown in Fig.2. For this purposes, the Cinema 4D studio and Photoshop were used. The next step of the 3D object modeling was optimization. The model inequalities were removed, the holes produced during the reconstruction were filled, the whole number of the polygons was reduced and the mesh grid was optimized. The mesh grid optimization was necessary to apply because of the correct model creation especially around the head of the animal. During this step,

model rotation and symmetry was also improved. These steps resulted in the creation of the four different 3D models of mammals, namely wolf, brown bear, fox and deer. These models will serve as testing database and the results will be compared with the results from the real images.

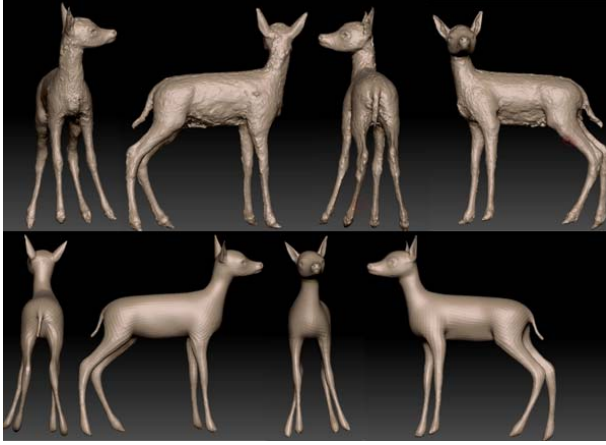


Fig. 2. The optimization of the 3D model

3. OBJECT RECOGNITION PROCESS

Object recognition process is shown in Fig.3 and can be divided into two parts: training and testing. Task of training part is to create a classification model from the training data. Training data contain a collection of images from each class. The first step is the feature extraction. In our work, visual descriptors like SIFT (Scale Invariant Feature Transform) and SURF (Speeded Up Robust Features) were chosen for this task. SIFT and SURF are well known descriptors, which are widely used in the object classification systems. More information about descriptors can be found in [4-6]. Moreover, the features extracted from images are used to create the classification model.

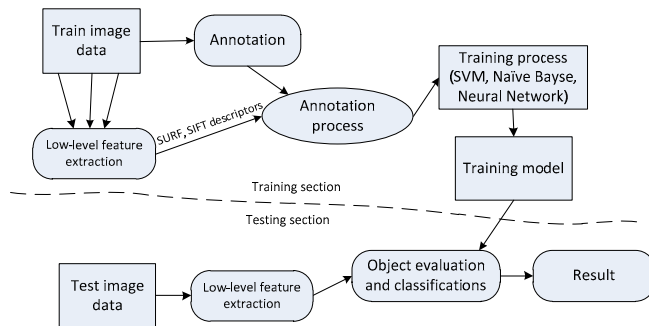


Fig. 3. Object recognition process

In the input of the testing part are the images with unknown animal objects designated for the classification. Moreover, these objects have the same metadata description like data in training part. Based on the metadata description, the classifier is able to successfully evaluate the unknown objects to the appropriate class.

4. CLASSIFICATION MODEL CREATION

In our work, combination of the bag the visual key points (BOW) method and the Support Vector Machine (SVM) were used. BOW is based on the quantization of the affine invariant visual descriptors and was presented in [7]. SVM is widely used in the image processing and object classification [8, 9]. SVM is non-probabilistic binary linear classifier and belongs to the group of model based classifiers. Its main task is to find an optimal hyperplane between two classes with maximum margin. In the model creation, the radial basic function (RBF) kernel was used.

5. EXPERIMENTAL RESULTS

The experiments were done using C++ programming language with OpenCV (Open Computer Vision) libraries. The training database consists of 5 classes: wild boar, wolf, fox, deer and brown bear. The number of the images in the training database pre class was from 50 to 100 images. The images from the training classes are shown in Fig. 4.



Fig. 4. The images from the training database

There were two different testing databases. The first one was created by randomly selected and removed images from training database and consists of 50 images, 10 per class. The second testing database was created from the 3D models of the animal, where these models were inserted to the background images. The second database also consists of 50 images, 10 per class. The examples from the second testing database are show in Fig.5. In the 3D object testing database, the images of the wild boar are the same as in the first testing database. We weren't able to create the 3D object of the wild boar, because we didn't have any model of the mounted wild board.

The experiments were done according to the object recognition process shown in Fig.3. There were tested more standalone runs, where one run consists of the combination of key point detector, key point descriptor, descriptor matcher in BOW vocabulary creation and during the SVM training. Four detectors, SURF, SUSIFT (SURfSIFT), SIFT and SISURF (SiftSURF), four descriptors SURF, OpponentSURF, SIFT, OpponentSIFT and two feature matcher, BruteForce (BF) and FlannBased (FB) were used.

SUSIFT a SISURF are hybrid key points detectors and was presented in [10]. The number of the descriptors was changing from 10000 to 20000 per class with step 2000 in BOW vocabulary creation and SVM training.

The experiments were done for both testing databases with the same training database. Then, the success rate of object classification for the same combination was compared and best results are given.



Fig. 5. The images from the training database

The best classification results for SIFT descriptor are shown in Fig. 6. The horizontal axis defines the keypoint detector and the vertical axis defines success rate of classification. Only the best result for every keypoint detector for the both databases, real animal objects and 3D modeled objects.

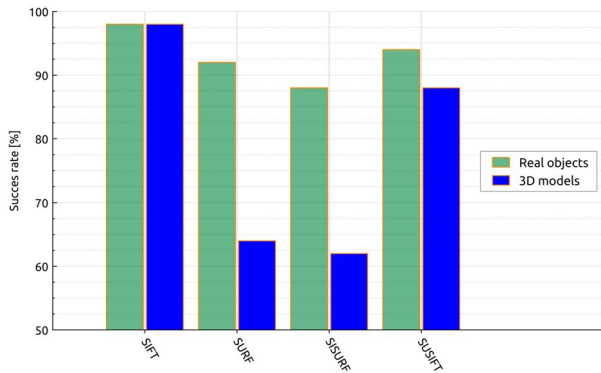


Fig. 6. Results for SIFT descriptor

The best success rate for SIFT descriptor at 98% was achieved in combination with SIFT detector. The 3D models classification achieved comparable success rate in the case of SIFT and SUSIFT detector, but in the combination with

SURF and SISURF detector was success rate notably lower, only around 64 %.

The best classification results for SURF descriptor are shown in Fig. 7. The horizontal axis defines the keypoint detector and the vertical axis defines success rate of classification. The best success rate for SURF descriptor at 90% was achieved in combination with SURF detector. The 3D objects classification achieved almost the same results as the real objects and in combination with SIFT detector the success rate was even higher.

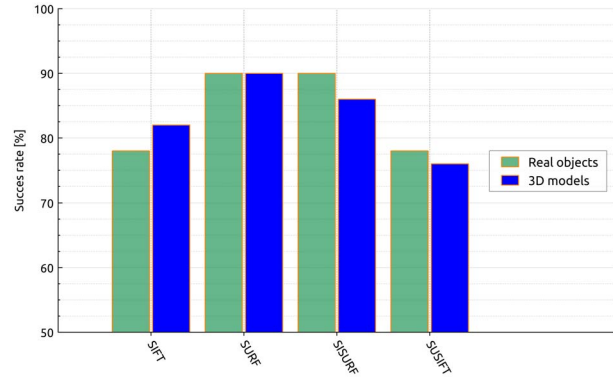


Fig. 7. Results for SURF descriptor

The best classification results for OpponentSIFT descriptor are shown in Fig. 8. The horizontal axis defines the keypoint detector and the vertical axis defines success rate of classification. From the given figure is evident, that OpponentSIFT descriptor achieved the best results in the combination with the all type of the descriptors at 98%. In the case of the SIFT and SUSIFT detectors the difference between real objects and 3D models classification success rate was minimal or none. In the case of SURF and SISURF descriptors there was significant difference between two databases.

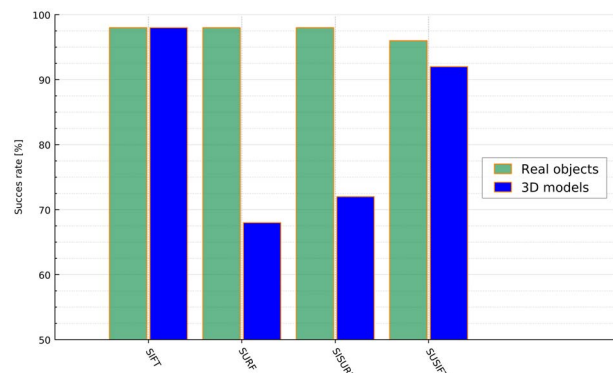


Fig. 8. Results for OpponentSIFT descriptor

The best classification results for OpponentSURF descriptor are shown in Fig. 9. The horizontal axis defines the keypoint detector and the vertical axis defines success rate of classification. The results from the two different databases

are almost the same in combination with the all type of descriptors for OpponentSURF descriptor. The best success rate was achieved in combination with SISURF detector at 96% for the database of the 3D models. In the combination with SURF detector, the best success rate was at 98%.

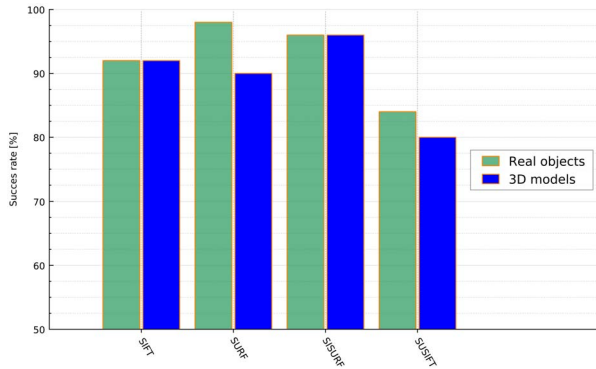


Fig. 9. Results for OpponentSURF descriptor

The result summary for the real objects is shown in the Table I. For the best combination for particular descriptor, there are shown the results on the both databases, the real objects and the 3D models.

TABLE I. THE BEST RESULTS FOR PARTICULAR DESCRIPTORS ON THE REAL OBJECTS

Classification model	Real objects Success rate[%]	3D models Success rate[%]
SIFT, SIFT, BF, 16000, 18000	98	94
SISURF, SURF, FB, 20000, 16000	90	84
SURF, OpponentSIFT, BF, 20000, 20000	98	72
SURF, OpponentSURF, 20000, 16000	98	80

TABLE II. THE BEST RESULTS FOR PARTICULAR DESCRIPTORS ON THE 3D MODELS

Classification model	Real objects Success rate[%]	3D models Success rate[%]
SIFT, SIFT, BF, 14000, 12000	90	98
SURF, SURF, FB, 18000, 20000	88	90
SIFT, OpponentSIFT, FB, 20000, 14000	96	98
SISURF, OpponentSURF, 16000, 12000	94	96

The result summary for the 3D objects is shown in the Table II. For the best combination for particular descriptor, there are shown the results on the both databases, the 3D models and the real objects.

6. CONCLUSION AND FUTURE WORK

In this paper, a new way of the testing database creation based on 3D modeling was presented. The 3D models were created using photogrammetry and 3D modeling for ideal 3D models of mammals. The 3D model database consists of four different modeled mammals, brown bear, fox, deer and wolf. This database served as the testing database in the object recognition process and the results were compared with database with the real objects. From the realized experiment is evident, that the success rate of the 3D model database was in the most case comparable with success rate of the database with the real objects.

According to our experiments, the testing database can be replaced with the 3D model database. Creating the testing database using 3D modeling can save a lot of time and money and we can model the different scenes and situations.

In the future, the 3D models will be animated and the videos with 3D models will be created. With these videos, we can simulate the migration of the wild animals and then we can create the migration corridors for particular mammals in particular areas. After the successful simulation, we will deploy the system of the automatic animal recognition from video sequences into the real world.

7. REFERENCES

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